The Binomial Theorem

ANSWER KEY

Section 1: Pascal's Triangle and Combinations

1 State the formula for the combination $\binom{n}{r}$.

a. $\frac{n!}{r!\,(n-r)!}$ ✓

b. $\frac{n!}{(n-r)!}$

c. $\frac{n!}{r!}$

d. $n \times r$

$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$ gives the coefficient of the $(r+1)$-th term in $(a+b)^n$.

2 Evaluate $\binom{6}{2}$.

a. 15 ✓

b. 30

c. 12

d. 20

$\binom{6}{2} = \frac{6 \times 5}{2 \times 1} = 15$.

3 What is row 4 of Pascal's Triangle (corresponding to $n = 4$)?

a. 1, 4, 6, 4, 1 ✓

b. 1, 3, 3, 1

c. 1, 5, 10, 10, 5, 1

d. 1, 4, 8, 4, 1

Row 4 is $\binom{4}{0} = 1$, $\binom{4}{1} = 4$, $\binom{4}{2} = 6$, $\binom{4}{3} = 4$, $\binom{4}{4} = 1$.

Section 2: Expanding Binomials

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4 State the Binomial Theorem expansion for $(a+b)^n$.

a. $\sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$ ✓

b. $\sum_{r=0}^{n} a^{n-r} b^r$

c. $\sum_{r=1}^{n} \binom{n}{r} a^r b^{n-r}$

d. $\sum_{r=0}^{n} n!\, a^r b^r$

The Binomial Theorem states $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$.

5 Expand $(x+2)^4$.

a. $x^4 + 8x^3 + 24x^2 + 32x + 16$ ✓

b. $x^4 + 4x^3 + 6x^2 + 4x + 16$

c. $x^4 + 16$

d. $x^4 + 8x^3 + 16x^2 + 32x + 16$

$\binom{4}{0}x^4 + \binom{4}{1}x^3(2) + \binom{4}{2}x^2(4) + \binom{4}{3}x(8) + \binom{4}{4}(16) = x^4 + 8x^3 + 24x^2 + 32x + 16$.

6 Find the first three terms in the expansion of $(2x-1)^5$ in descending powers of x.

a. $32x^5 - 80x^4 + 80x^3$ ✓

b. $32x^5 - 16x^4 + 8x^3$

c. $32x^5 + 80x^4 + 80x^3$

d. $32x^5 - 40x^4 + 20x^3$

Term 1: $\binom{5}{0}(2x)^5 = 32x^5$. Term 2: $\binom{5}{1}(2x)^4(-1) = 5(16x^4)(-1) = -80x^4$. Term 3: $\binom{5}{2}(2x)^3(-1)^2 = 10(8x^3)(1) = 80x^3$.

Section 3: Finding Specific Terms

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7 State the formula for the $(r+1)$-th term in the expansion of $(a+b)^n$.

a. $T_{r+1} = \binom{n}{r} a^{n-r} b^r$ ✓

b. $T_{r+1} = \binom{n}{r+1} a^{n-r} b^r$

c. $T_{r+1} = \binom{n}{r} a^r b^{n-r}$

d. $T_{r+1} = a^{n-r} b^r$

The general term is given by $T_{r+1} = \binom{n}{r} a^{n-r} b^r$ where r indexes from 0 to n.

8 Find the 4th term in the expansion of $(x+3)^7$.

a. $945x^4$ ✓

b. $315x^4$

c. $2835x^3$

d. $105x^5$

For the 4th term, $r=3$. $T_4 = \binom{7}{3} x^{7-3}(3)^3 = 35 \cdot x^4 \cdot 27 = 945x^4$.

9 Find the coefficient of x^3 in the expansion of $(2x-3)^6$.

a. -4320 ✓

b. 4320

c. -1440

d. 2160

We want x^3, so power of $(2x)$ is 3, which means $n-r=3 \Rightarrow r=3$. Term is $\binom{6}{3}(2x)^3(-3)^3 = 20(8x^3)(-27) = -4320x^3$.

Section 4: Term Independent of x and Sum of Coefficients

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10 Find the term independent of x (the constant term) in the expansion of $\left(x+\frac{2}{x}\right)^6$.

a. 160 ✓

b. 80

c. 64

d. 20

General term: $\binom{6}{r}x^{6-r}(2x^{-1})^r=\binom{6}{r}2^r x^{6-2r}$. Set exponent to 0: $6-2r=0\Rightarrow r=3$. Constant term $=\binom{6}{3}2^3=20\times 8=160$.

11 What is the sum of all coefficients in the expansion of $(2x-3y)^4$?

a. 1 ✓

b. -1

c. 625

d. 81

To find the sum of coefficients, substitute $x=1$ and $y=1$: $(2(1)-3(1))^4=(-1)^4=1$.

12 What is the sum of the elements in row n of Pascal's Triangle (i.e. $\sum_{r=0}^{n}\binom{n}{r}$)?

a. 2^n ✓

b. n^2

c. $2n$

d. 2^{n-1}

By the Binomial Theorem with $a=1, b=1$: $(1+1)^n=2^n=\sum_{r=0}^{n}\binom{n}{r}$.

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